Lecture 19: Max-Flow Min-Cut

Jessica Sorrell

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601.433/633 Introduction to Algorithms
Slides by Mike Dinitz

Introduction

Flow Network:

- Directed graph $G = (V, E)$
- *Capacities* $c : E \to \mathbb{R}_{\geq 0}$ (simplify notation: $c(x, y) = 0$ if $(x, y) \notin E$)
- Source $s \in V$, sink $t \in V$

Today: flows and cuts

- Flow: "sending stuff" from s to t
- Cut: separating t from s

Turn out to be very related!

Today: some algorithms but not efficient. Mostly structure. Better algorithms Tuesday.

Flows

Intuition: send "stuff" from s to t

- Water in a city water system, traffic along roads, trains along tracks, ...

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for all $v \in V \setminus \{s, t\}$. This constraint is known as *flow conservation*.

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$$|f| = \sum_{u:(s,u)\in E} f(s,u) - \sum_{u:(u,s)\in E} f(u,s) = \sum_{u:(u,t)\in E} f(u,t) - \sum_{u:(t,u)\in E} f(t,u)$$

Feasible Flows

Capacity constraints: $\mathbf{0} \leq \boldsymbol{f}(\boldsymbol{u}, \boldsymbol{v}) \leq \boldsymbol{c}(\boldsymbol{u}, \boldsymbol{v})$ for all $(\boldsymbol{u}, \boldsymbol{v}) \in \boldsymbol{V} \times \boldsymbol{V}$

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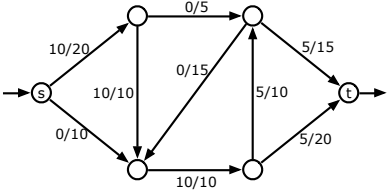
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- If $f(e) = c(e)$ then f *saturates* e.
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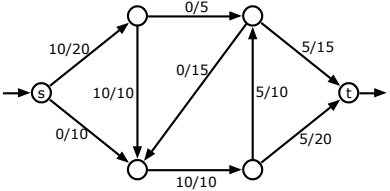
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Problem we'll talk about: find feasible flow of maximum value (max flow)

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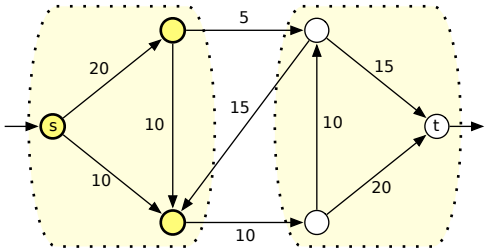
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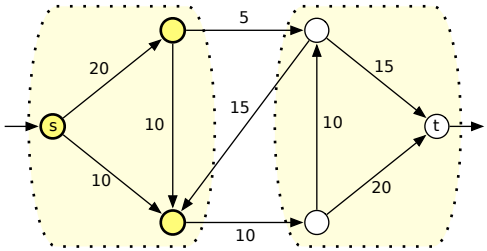


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Problem we'll talk about: find (s, t)-cut of minimum capacity (min cut)

Warmup Theorem

Theorem

Let $\boldsymbol{f}$ *be a feasible* $(\boldsymbol{s}, \boldsymbol{t})$*-flow, and let* $(\boldsymbol{S}, \bar{\boldsymbol{S}})$ *be an* $(\boldsymbol{s}, \boldsymbol{t})$*-cut. Then* $|\boldsymbol{f}| \leq \boldsymbol{cap}(\boldsymbol{S}, \bar{\boldsymbol{S}})$.

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$$|f| = \sum_{v \in V} f(s,v) - \sum_{v \in V} f(v,s) \qquad \text{(definition)}$$

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&\leq \sum_{u \in S} \sum_{v \in \bar{S}} c(u,v) = cap(S,\bar{S}) && \text{(flow is feasible)}
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Max-Flow Min-Cut

Corollary

If $\boldsymbol{f}$ avoids every $\bar{\boldsymbol{S}} \to \boldsymbol{S}$ edge and saturates every $\boldsymbol{S} \to \bar{\boldsymbol{S}}$ edge, then $\boldsymbol{f}$ is a maximum flow and $(\boldsymbol{S}, \bar{\boldsymbol{S}})$ is a minimum cut.

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Theorem (Max-Flow Min-Cut Theorem)

In any flow network, value of max (s,t)-flow = capacity of min (s,t)-cut.

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Theorem (Max-Flow Min-Cut Theorem)

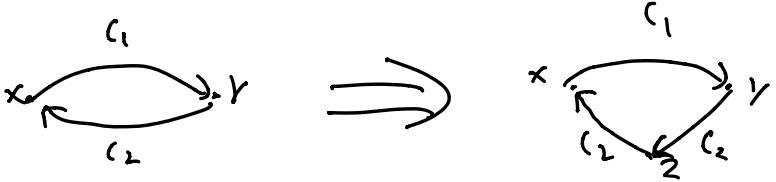
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Spend rest of today proving this.

- Many different valid proofs.
- We'll see a classical proof which will naturally lead to algorithms for these problems.

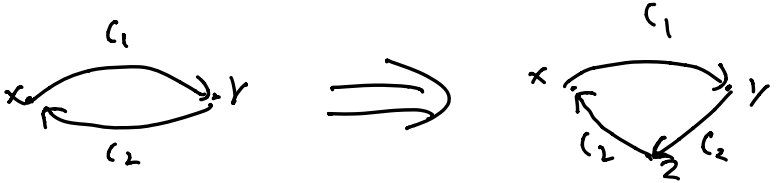
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- Doesn't change max-flow or min-cut
- Increases #edges by constant factor, # nodes by at most original # edges.

Residual

Let f be feasible (s, t)-flow. Define *residual capacities*:

$$c_f(u, v) = \begin{cases} c(u, v) - f(u, v) & \text{if } (u, v) \in E \\ 0 & \text{otherwise} \end{cases}$$

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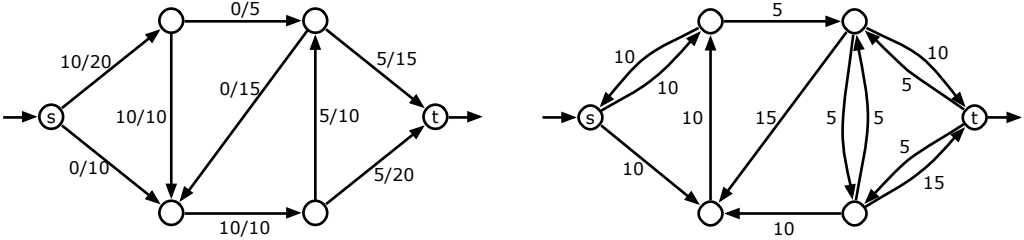
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Residual Graph: $G_f = (V, E_f)$ where $(u,v) \in E_f$ if $c_f(u,v) > 0$.



A flow f in a weighted graph G and the corresponding residual graph G_f.

Start of Proof

Let f be a max (s, t)-flow with residual graph G_f.
Want to Show: There is a cut $(S, \bar{S})$ with $cap(S, \bar{S}) = |f|$.

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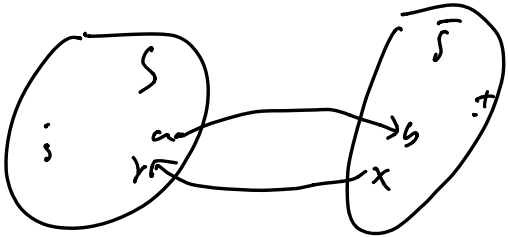
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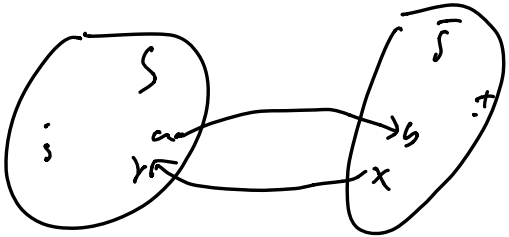


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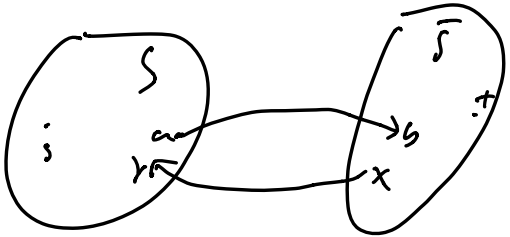
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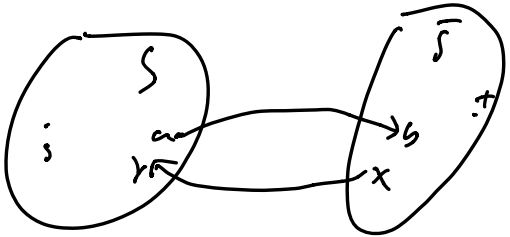
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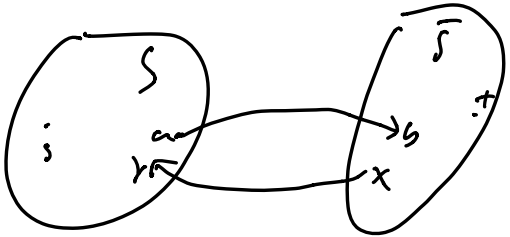
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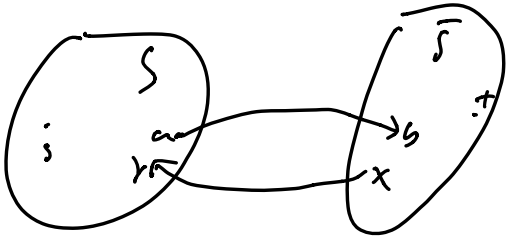
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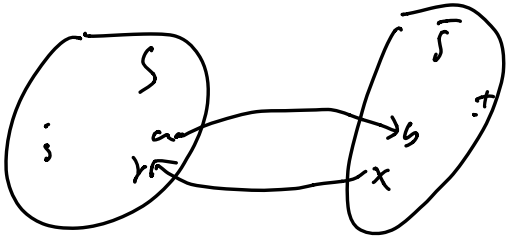
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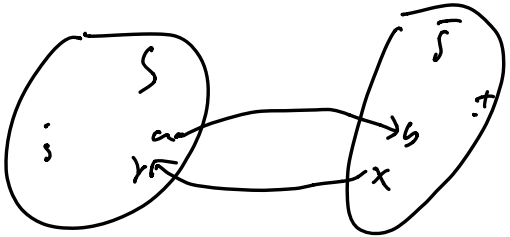
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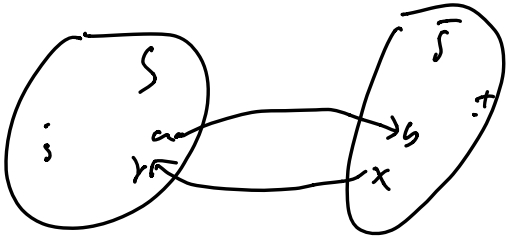
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f saturates $S \to \bar{S}$ edges, avoids $\bar{S} \to S$ edges $\implies cap(S, \bar{S}) = |f|$ by corollary

Case 2

Suppose $\exists$ an $s \to t$ path P in G_f.

- Called an *augmenting path*

Idea: show that we can "push" more flow along P, so f not a max flow. Contradiction, can't be in this case.

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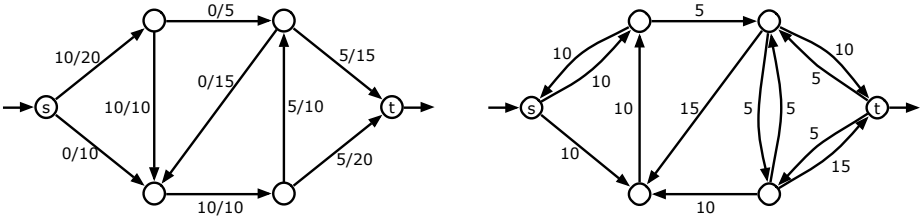
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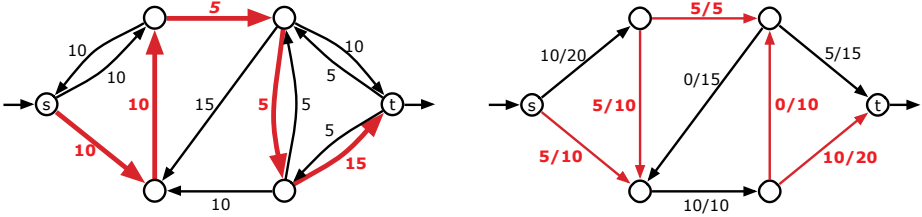
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- Foreshadowing: augmenting path allows us to send more flow. Algorithm to increase flow!

Intuition



A flow f in a weighted graph G and the corresponding residual graph G_f.



An augmenting path in G_f with value $F = 5$ and the augmented flow f'.

Formalities

Let P be (simple) augmenting path in G_f. Let $F = \min_{e \in P} c_f(e)$.

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Define new flow f': for all $(u, v) \in E$, let

$$f'(u,v) = \begin{cases} f(u,v) + F & \text{if } (u,v) \text{ in } P \\ f(u,v) - F & \text{if } (v,u) \text{ in } P \\ f(u,v) & \text{otherwise} \end{cases}$$

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$$f'(u,v) = \begin{cases} f(u,v) + F & \text{if } (u,v) \text{ in } P \\ f(u,v) - F & \text{if } (v,u) \text{ in } P \\ f(u,v) & \text{otherwise} \end{cases}$$

Claim: f' is a feasible (s,t)-flow with $|f'| > |f|$.

Formalities

Let P be (simple) augmenting path in G_f. Let $F = \min_{e \in P} c_f(e)$.

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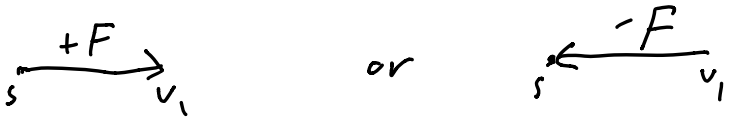
Plan: prove (sketch) each subclaim individually

- $|f'| > |f|$
- f' an (s,t)-flow (flow conservation)
- f' feasible (obeys capacities)

$|f'| > |f|$

Consider first edge of P (out of s), say (s, v_1)

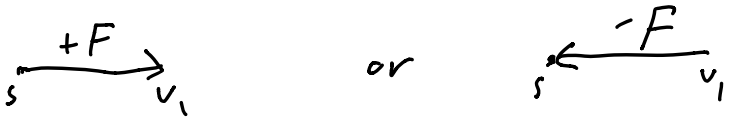
- If $(s, v_1) \in E$, then $f'(s, v_1) = f(s, v_1) + F$
- If $(v_1, s) \in E$ then $f'(v_1, s) = f(v_1, s) - F$



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$$|f'| = \sum_u f'(s, u) - \sum_u f'(u, s) = |f| + F > |f|$$

f' obeys flow conservation

Consider some $u \in V \setminus \{s, t\}$.

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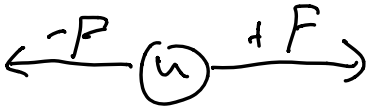
- If $u \notin P$, no change in flow at $u \implies$ still balanced.

f' obeys flow conservation

Consider some $u \in V \setminus \{s, t\}$.

- If $u \notin P$, no change in flow at $u \implies$ still balanced.
- If $u \in P$, four possibilities:









f' obeys capacity constraints

Let $(u, v) \in E$

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- If $(u, v), (v, u) \notin P$: $f'(u, v) = f(u, v) \leq c(u, v)$

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$$
\begin{aligned}
f'(u, v) &= f(u, v) + F \\
&\leq f(u, v) + c_f(u, v) \\
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  - If $(v, u) \in P$:

$$\begin{aligned} f'(u, v) &= f(u, v) - F \\ &\geq f(u, v) - c_f(v, u) \\ &= f(u, v) - f(u, v) \\ &= 0 \end{aligned}$$

Ford-Fulkerson Algorithm and Integrality

FF Algorithm

Obvious algorithm from previous proof: keep pushing flow!

$f = \vec{0}$
while($\exists s \to t$ path P in G_f) {
  $F = \min_{e \in P} c_f(e)$
  Push F flow along P to get new flow f'
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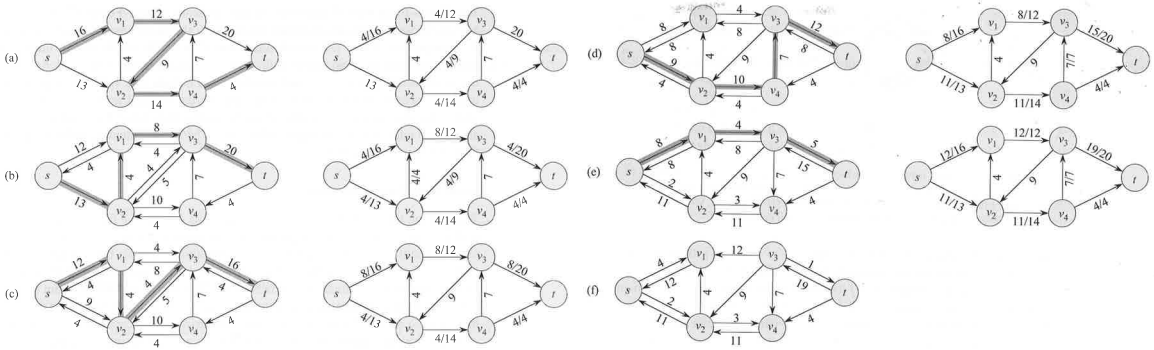
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Correctness: directly from previous proof

Example



Integrality

Corollary

If all capacities are integers, then there is a max flow such that the flow through every edge is an integer

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Corollary

If all capacities are integers, then there is a max flow such that the flow through every edge is an integer

Proof.

Induction on iterations of the Ford-Fulkerson algorithm: initially true, stays true $\implies$ true at end. □

Running Time

Theorem

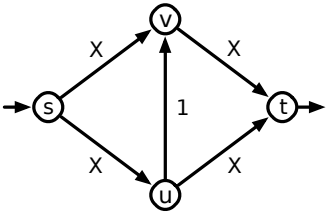
If all capacities are integers and the max flow value is F, Ford-Fulkerson takes time at most $O(F(m+n))$

Running Time

Theorem

If all capacities are integers and the max flow value is F*, Ford-Fulkerson takes time at most* $O(F(m+n))$

Finding path takes $O(m+n)$ time, increase flow by at least 1



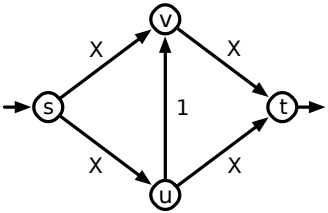
A bad example for the Ford-Fulkerson algorithm.

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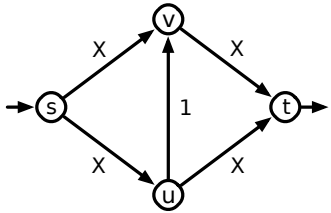
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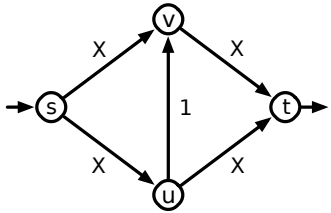
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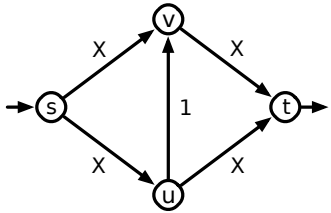
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$\Longrightarrow$ Exponential time!